

Passive RF-Silent Drone Detection via Neuromorphic Vision and Acoustic Fusion of the Propeller Blade-Pass Frequency

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Abstract—We investigate passive, RF-silent drone detection by exploiting the fact that a spinning propeller emits a characteristic blade-pass frequency (BPF) simultaneously as optical flicker, sensed by a neuromorphic event camera, and as acoustic pressure, sensed by a microphone array. To our knowledge no prior work fuses these two modalities for counter-UAS. We present a working prototype that extracts a per-pixel flicker frequency from a Prophesee EVK4-HD event camera, isolates propeller regions by connected-component clustering and temporal tracking, extracts the acoustic BPF and a direction-of-arrival estimate from a four-microphone array, and fuses the two with a sequential Bayesian updater using harmonic-aware frequency cross-validation. The central finding is methodological: detection *architecture*, not parameter tuning, determines success. A grid-FFT baseline failed at range regardless of tuning, whereas per-pixel frequency extraction with connected-component clustering gave stable, zero-false-positive detection with first-detect latency reduced from 600–1200 ms to ~ 100 ms after real-time optimization. Cross-modal frequency agreement worked reliably; cross-modal *spatial* calibration did not converge indoors ($R^2 = 0.14$), which we report as an instructive negative result.

Index Terms—event camera, neuromorphic vision, drone detection, sensor fusion, blade-pass frequency, Bayesian inference

I. INTRODUCTION

Unauthorized consumer drones are now an operational threat: the 2018 Gatwick incident diverted ~ 1000 flights, and drones flying pre-programmed GPS missions emit no radio signal, making them invisible to RF scanning—the dominant commercial technology. Since only a few US federal agencies may lawfully jam a drone, most operators are restricted to *detection*, motivating passive sensing. Every existing modality has a structural blind spot (Table I). Prior counter-UAS work spans RF fingerprinting [4], [5] and acoustic signature detection with array localization [6], [7]; we are not aware of any that fuses an event camera with a microphone array.

Hypothesis. The BPF appears in both the optical and acoustic domains; requiring *cross-modal frequency agreement* suppresses false-alarm sources that live in only one modality (flickering luminaires are silent; HVAC does not flicker), driving the joint false-positive probability down.

Contributions. (i) A prototype—to our knowledge the first—fusing an event camera with a microphone array for drone detection via the shared BPF. (ii) An architectural result:

TABLE I
STRUCTURAL WEAKNESSES OF EXISTING DETECTION MODALITIES

Modality	Structural weakness
RF scanning	Blind to autonomous / RF-silent drones
Radar	Confuses small drones with birds; costly
RGB / thermal	Motion blur, ~ 60 dB range, low-light/fog failure
Acoustic only	Short range; 10–20 % false positives in noise

per-pixel frequency plus connected-component clustering succeeds where grid-FFT fails at range regardless of tuning. (iii) Harmonic-aware cross-modal frequency validation and gated Bayesian fusion. (iv) Real-time engineering cutting first-detect latency $6\text{--}12\times$. (v) An honest account of negative results.

II. PHYSICAL PRINCIPLE

An event camera reports asynchronous per-pixel brightness-change events $(x, y, t, \text{polarity})$ with microsecond timing and >120 dB dynamic range [1]–[3]; a 30 fps camera samples a 200 Hz BPF below the Nyquist rate and blurs the blades, whereas an event camera measures the per-pixel oscillation directly. A propeller with B blades at shaft rate f_s modulates both light and sound at

$$f_{\text{BPF}} = Bf_s = B \text{ RPM}/60, \quad (1)$$

typically 80–300 Hz for common multirotors (1–2.5 kHz for racing quads). The signature is unavoidable, narrowband, and rare in nature, so requiring the two channels to agree (up to harmonic ambiguity, Section III-C) rejects single-modality noise.

III. METHOD

A. Visual Pipeline

Grid-FFT baseline (failed). Dividing the 1280×720 sensor into a 16×16 grid, counting events per cell, and running a per-cell FFT detected the propeller only at centimeter range. At distance a propeller’s few pixels are diluted by hundreds of noise pixels in the same cell; relaxing thresholds merely promoted noise to 25–43 false “regions” per frame. *Tuning cannot fix a wrong architecture.*

Per-pixel frequency + connected components. We instead threshold the SDK per-pixel frequency map to a binary mask, dilate, run connected-component labeling, and keep clusters of ≥ 2 pixels whose frequency coefficient of variation is ≤ 0.3 . Clustering adapts to the propeller’s true shape—even 2–3 consistent pixels are meaningful—so there is no signal dilution. A single-pixel minimum produced 13–17 false detections per frame, confirming that spatial coherence (≥ 2 connected pixels) is the primary false-positive filter. A temporal tracker with greedy association (spatial < 100 px, frequency $< 30\%$) and velocity-predictive matching confirms only persistent tracks. On the bench propeller this gave one stable track at 270 Hz BPF (5400 RPM, 3 blades), locked position, and zero false positives.

Real-time optimization. Full-resolution morphology and connected components at 25 Hz cost ~ 315 ms/s of CPU, causing growing lag. A $4\times$ downscale before morphology (statistics still on full-resolution regions of interest), in-place drawing, and early-exit on empty masks removed all drift. A production pass then pre-allocated all buffers (eliminating garbage-collection jitter) and replaced hard hit-count confirmation with quality-weighted Bayesian confidence: each observation contributes $p = 0.65 + q(0.85 - 0.65)$, with q from pixel count and frequency consistency, accumulated as $P = 1 - (1 - p)^n$, decayed when idle, confirming at $P \geq 0.95$. Strong detections then confirm in 2 frames and weak ones in 4, preserving zero false positives.

B. Acoustic Pipeline

A rolling 0.2 s window per channel is Hann-windowed and transformed; the four magnitude spectra are incoherently averaged (~ 6 dB gain) and the peak picked in the target band with an SNR gate. Direction of arrival uses frequency-weighted GCC-PHAT [8] per opposing 64 mm mic pair with $4\times$ zero-padded sub-sample delay, parabolic interpolation, a peak/second-peak confidence gate, and a median filter, giving usable azimuth on steady tones ($\sim \pm 10^\circ$; the planar array cannot resolve elevation). ODAS [9] and the array’s speech-tuned firmware DOA both failed on impulsive propeller acoustics, motivating the custom estimator.

C. Fusion

One sensor often reports a harmonic of the other, so we search pairs (n_v, n_a) , $1 \leq n_v, n_a \leq 4$, and declare a match when $|f_v/n_v - f_a/n_a|/(f_a/n_a) < 0.05$ with confidence $1/(n_v n_a)$. A sequential Bayesian detector starts at prior $P = 0.001$, updates per modality with sigmoid likelihoods, adds a frequency-match bonus, and decays toward the prior absent evidence; detection fires at $P > 0.8$. Quality, temporal-association (≤ 200 ms), and angle-residual ($\leq 35^\circ$) gates prevent overconfident fusion. In a live run with no propeller in view the detector latched on strong acoustic tones via the audio-only path and decayed when they stopped—single-modality dominance is a real failure mode the gates must police.

TABLE II
VISUAL DETECTOR PERFORMANCE, v2 vs. v3

Metric	v2	v3	Gain
First detect (strong)	600–1200 ms	~ 100 ms	$6\text{--}12\times$
First detect (weak)	600–1200 ms	~ 200 ms	$3\text{--}6\times$
Steady-state update	100–140 ms	50–90 ms	$\sim 2\times$
Per-frame analysis	~ 1.5 ms	~ 0.8 ms	$\sim 2\times$
GC jitter	± 5 ms	$< \pm 1$ ms	$\sim 5\times$

IV. RESULTS

The headline result is latency (Table II): first-detect falls from 600–1200 ms to ~ 100 ms with jitter reduced $\sim 5\times$. The final configuration tracked multi-pixel clusters over 80+ consecutive hits with zero false positives on ~ 30 s recordings, and cross-modal frequency agreement fired reliably.

Azimuth calibration (negative result). Fitting a linear visual→acoustic azimuth map by sweeping a co-located source yielded $R^2 = 0.14$ (RMSE 0.70°): the acoustic DOA spanned only $\sim \pm 1.2^\circ$ against a $\sim \pm 25^\circ$ visual sweep, consistent with reverberation or an insufficient sweep. Cross-modal *spatial* fusion is therefore unvalidated; cross-modal *frequency* fusion is validated. Projected ranges ($\sim 50\text{--}150$ m, acoustic-limited) and the sub-0.1 % false-positive target are design-derived, pending outdoor validation.

V. DISCUSSION AND LIMITATIONS

Three lessons generalize. *Architecture beats tuning*: the decisive change was replacing grid-FFT counting with per-pixel frequency plus clustering. *Spatial coherence is the false-positive filter*: two connected consistent pixels are meaningful, one is not. *Probabilistic beats deterministic confirmation*: quality-weighted Bayesian confidence confirms strong detections $3\text{--}6\times$ faster than a hard hit count at the same false-positive rate.

Limitations: all detection results are indoor, single-source, ~ 30 s recordings; the planar array cannot resolve elevation; harmonic ambiguity (2-blade at 100 Hz $\equiv$ 4-blade at 50 Hz) needs external metadata; and there is no hardware inter-sensor clock (~ 100 ms association uncertainty). Planned work: a headless C++ port for deterministic sub-millisecond timing, a non-planar array for elevation, hardware synchronization, multi-target tracking, and outdoor calibration with a wider sweep. The validated contributions stand independently of the range and false-positive projections, which await outdoor validation on labeled data.

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